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RESEARCH ARTICLE

The Effect of AI Gamification on Students' Engagement and Academic Achievement in Malaysia: SEM Analysis Perspectives

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ABSTRACT Gamification and Artificial Intelligence (AI) have transformed the educational environment, offering innovative approaches to enhance student engagement and learning outcomes. By using the power of gamification and AI, educators can create immersive, personalized, and interactive learning experiences that increase motivation, participation, and academic achievement. This study investigates the relationships between gamification, artificial intelligence, student engagement, and achievement in educational contexts. A quantitative approach was employed, combining online surveys and structural equation modeling (SEM) analysis with 290 students from Malaysia. SEM analysis was performed using SmartPLS 4 software. The findings reveal significant positive relationships between gamification and student achievement, gamification and student engagement, gamification and artificial intelligence, and AI with both student engagement and achievement. Furthermore, student engagement was found to influence student achievement positively. The study's implications suggest that educators and instructional designers should adopt these technologies to create intelligent, interactive, and personalized learning environments. Future research directions include exploring long-term effects and contextual factors influencing student outcomes to promote student success in the digital age. However, the study is limited to a single country context (Malaysia), and relies on self-reported survey data, which may influence generalizability and introduce potential biases.

INDEX TERMS Artificial intelligence, education, gamification, human-AI collaboration, personalized learning, student engagement, technology-enhanced learning.

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I. INTRODUCTION

The recent advancement of technology over the past few decades has significantly reshaped the educational

landscape. Modern educational paradigms now integrate virtual collaboration, mobile learning applications, and technology-enhanced learning programs to bolster student outcomes [1], [2], [3], [4]. Such innovations have fueled interest in alternative classroom structures, including augmented classrooms and blended learning environments, which have demonstrated potential for improving educational engagement and effectiveness [5], [6], [7], [8]. However, logistical constraints and resource limitations often hinder the widespread implementation of these approaches, necessitating the exploration of more accessible and scalable educational innovations [3], [9].

Traditional educational systems face several critical challenges, such as an over-reliance on standardized testing and passive learning methods, contributing to student disengagement and anxiety [10], [11]. Teachers often struggle to incorporate 21st-century teaching approaches due to a lack of knowledge and training, further compounding these issues [12], [13]. Additionally, outdated textbooks and insufficient resources, including limited access to technology and modern teaching materials, constrain the ability of educators to provide an engaging and relevant learning experience [14]. These challenges highlight the need for innovative pedagogical strategies that can be easily implemented and positively impact student motivation and academic achievement [15], [16].

One such promising strategy is gamification, which involves applying game-design elements and principles in non-game contexts to enhance user engagement and motivation [17], [18], [19]. Despite its growing popularity, the theoretical and psychological mechanisms underpinning gamification's effectiveness remain underexplored, particularly in educational settings [18], [20]. Research has begun to reveal how individual differences, such as personality behaviors and prior video game experience, can influence the outcomes of gamified learning interventions, underscoring the need for further investigation into these moderating factors [21], [22].

Studies have shown that gamification can significantly enhance student engagement, motivation, and academic performance. For instance, [23] found that gamification positively impacted students' learning, engagement, and behavior, with varying effects based on personality traits. Introverted students, for example, achieved higher points and badges, while extroverted students showed more significant improvements in accuracy. Conscientious students demonstrated higher accuracy and achievement, and agreeable students exhibited increased engagement. Similarly, Khaleel et al. [24] reported that gamified learning environments led to higher levels of student engagement and academic performance than non-gamified settings, suggesting that gamification can be a powerful tool for enhancing educational outcomes.

In parallel with gamification, artificial intelligence (AI) offers transformative potential in education through

personalized learning experiences [23]. AI-powered systems can analyze student data to identify strengths and weaknesses, tailoring learning paths to meet individual needs [25], [26], [27]. Integrating AI and gamification can create highly personalized, adaptive learning environments that enhance student motivation and engagement [28], [29]. These systems provide real-time feedback and support, helping to bridge learning gaps and promote deeper understanding and critical thinking skills [30], [31].

The combined use of gamification and AI in educational settings promises to address many of the shortcomings of traditional education [32], [33]. For example, AI-driven adaptive learning systems can reduce learning time and improve outcomes by continuously adjusting to students' progress and preferences [28], [31]. Furthermore, applying these technologies can foster student autonomy and self-directed learning and enhance collaboration and social interaction among students, leading to increased creativity and problem-solving abilities [34], [35].

Beyond education, the principles of gamification and AI can be applied to various fields, such as healthcare, marketing, and human resources, demonstrating their broad utility and impact [36], [37], [38]. In healthcare, gamification and AI can create engaging and personalized patient experiences that promote healthy behaviors and disease prevention [39], [40]. In marketing, AI-driven gamification can improve customer retention and loyalty through immersive and rewarding experiences [41]. In human resources, these technologies can enhance employee training and development, improving learning outcomes and knowledge retention [42], [43].

Moreover, integrating gamification and AI in education represents a significant opportunity to transform the learning experience, making it more personalized, engaging, and effective. As technological capabilities continue to advance, these innovations' potential applications and benefits will likely expand, offering new ways to address the challenges faced by traditional educational systems and beyond. This study aims to deepen our understanding of how gamification and AI can enhance student engagement, learning outcomes, and overall educational experiences, thereby contributing to the ongoing evolution of modern education.

This study explores the combined effects of gamification and AI-driven personalized learning on enhancing student engagement and academic performance in educational settings. Specifically, the study investigates how gamification and AI technologies influence students' learning experiences and outcomes, focusing on understanding the correlation between enhanced engagement and academic success by integrating these innovative approaches.

To achieve this aim, the study pursues the following objectives:

- To investigate how gamification and AI-driven personalized learning contribute to improving student engagement in educational contexts.

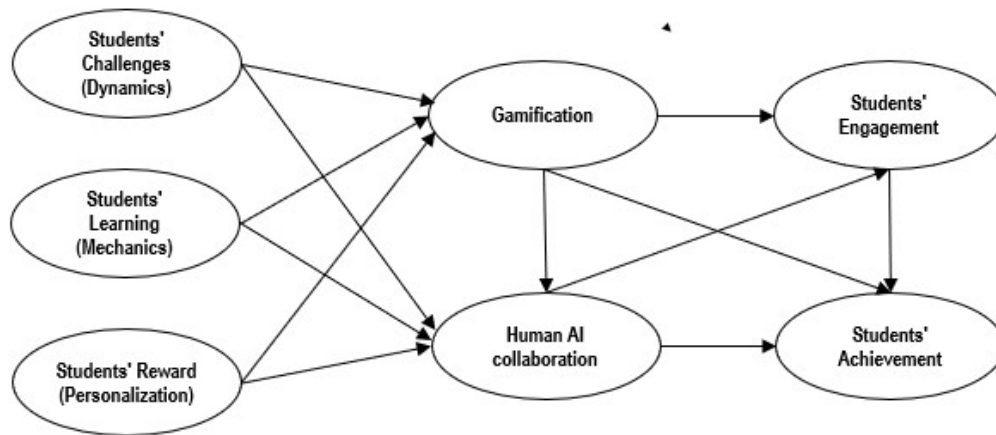


FIGURE 1. Proposed research model.

- To evaluate the effect of gamification and AI technologies on students' academic performance.
- To analyze the relationship between increased engagement and academic achievement within AI and gamified learning environments.

Based on these objectives, the study addresses the following research questions:

1. How do gamification and AI-driven personalized learning enhance student engagement?
2. What is the effect of gamification and AI technologies on academic achievement?
3. How does increased engagement relate to academic achievement in AI-enhanced gamified learning settings?

By addressing these questions, the study aims to broadly understand how these technologies interact to improve engagement and academic outcomes.

II. THEORETICAL BACKGROUND AND HYPOTHESIS DEVELOPMENT

Our proposed model integrates gamification and AI-driven personalized learning to enhance student engagement and academic achievement. This model is intended to address students' challenges (dynamics), learning processes (mechanics), and rewards (personalization), utilizing the combined effect between gamification and AI to create a more interactive and engaging learning environment, as shown in Figure 1. Furthermore, theoretical understanding and conceptual knowledge are discussed in subsequent sections.

Gamification is extensively researched in the education area in relation to enhancing learning and motivation. Various research studies indicate that incorporating game elements, including points and badges, enhances engagement and mental development [44], [45]. The current literature indicates disparate findings, prompting more scrutiny of its theoretical foundations [20]. Several theories are being studied to explain the mechanisms of gamification, such as activity theory [46], self-determination theory [47], situated learning

theory [48], situational relevance theory [49], achievement goal theory [50], and social cognitive theory [51]. Few other theories like cognitive load theory [52], the ARCS motivational model [53] and the technology acceptance model [54], Mechanics, dynamics and emotions model [44], [55] as important theoretical foundations in gamification research. On the other hand, review gamification in the educational context and emphasize "Lander's theory of gamified learning" as an important theoretical treatise in scientific studies, which includes self-determination theory, Game design theories, goal-setting theory, and behavior reinforcement theory [44].

Self-Determination Theory (SDT) and the MDA model have proven particularly useful to describe gamification's motivational and emotional impacts on learning environments among the various theoretical models. SDT, developed by Deci and Ryan [47], describes human motivation with intrinsic and extrinsic forms on an autonomy continuum. SDT comprises three universal psychological needs: competence, autonomy, and relatedness. Competence addresses performing well with the environment; autonomy addresses self-direction, and relatedness addresses bonding with others [47]. More autonomous motivation yields increased engagement and well-being, connecting SDT with gamification in learning [56]. Gamification supports SDT through game elements to fulfill these psychological needs. Feedback and leaderboards enhance competence through the accuracy of goals [57]. Choice in gamified contexts supports autonomy, and multiplayer mechanisms enhance relatedness [58]. Personalized learning with AI personalizes education according to individual needs, evoking intrinsic motivation and engagement.

The MDA model, developed by [59], categorizes game design into mechanics, dynamics, and aesthetics. Mechanics describe the rules of player action, dynamics involve player-system interaction, and aesthetics is concerned with emotional gameplay response [55]. This model supports

the effect of gamification on motivation and learning. Our model's mechanics are game elements such as points, badges, and challenges as extrinsic motivators based on behaviorist principles. Dynamics are interactions between the students and the elements and the AI-based learning system, enabling social learning and cooperation based on constructivist theory. Aesthetics are essential to induce positive affective states such as enjoyment and satisfaction, enhancing immersive learning experiences [44], [59].

Both theories, SDT and MDA, provide a general framework for analyzing the effects of gamification on learning. SDT illustrates how game mechanics satisfy psychological needs to promote intrinsic motivation, and MDA identifies structural and experiential features that maximize engagement. The integration captures gamified learning's motivational and design aspects, optimizing students' engagement and performance. Our model informs empirical research, ensuring gamification interventions increase engagement and address learners' motivational and cognitive demands with AI personalization.

A. STUDENTS' CHALLENGES (DYNAMICS) AND GAMIFICATION AND ARTIFICIAL INTELLIGENCE

Students in traditional educational systems often face multifaceted challenges that impede their academic engagement and learning outcomes [12], [60]. These challenges encompass passive learning approaches, lack of motivation, and standardized teaching methodologies that fail to accommodate diverse learning styles [12], [61]. Furthermore, limited access to technological resources and outdated instructional materials exacerbate the disconnect between students and their educational environment [62], [63]. Such dynamics underscore the need for innovative strategies to foster active engagement and cater to individualized learning needs [64].

Gamification, which involves incorporating game-like elements into educational settings, has emerged as a viable solution to enhance student engagement and motivation [65], [66]. Research suggests that gamified approaches can transform passive learning into an interactive and rewarding experience, encouraging active participation and fostering a sense of achievement [67], [68]. By addressing motivational deficits and creating a more stimulating learning environment, gamification offers a potential pathway for overcoming students' challenges [32], [69]. Therefore, we hypothesize:

Additionally, human-AI collaboration and personalized learning systems have gained traction as transformative tools to address individual learning needs [26], [70]. These approaches utilize advanced AI technologies to tailor educational content, adapt to diverse learning styles, and provide real-time feedback [71]. Personalized learning not only alleviates the one-size-fits-all issue but also empowers educators by streamlining administrative tasks and enabling a focus on student-centric teaching [72]. Adopting such systems is likely driven by the challenges students face as educators and institutions seek scalable solutions to enhance learning outcomes. Hence, we propose:

H1. Students' Challenges (Dynamics) will positively influence the implementation of gamification.

H2. Students' Challenges (Dynamics) will positively influence the adoption of human-AI collaboration/personalized learning.

These hypotheses aim to explore the role of educational challenges in driving innovation, mainly through gamification and human-AI collaboration, as potential solutions to enhance the learning experience.

B. STUDENTS' LEARNING (MECHANICS) AND GAMIFICATION AND ARTIFICIAL INTELLIGENCE

The mechanics of student learning encompass the processes and strategies that enable effective knowledge acquisition and skill development [59]. These mechanics are most impactful when they engage students actively, promote critical thinking, and foster problem-solving abilities [71]. However, traditional teaching methods often emphasize rote memorization and passive learning, which limit students' capacity for deeper understanding and practical application of knowledge [72]. To overcome these limitations, integrating interactive and dynamic elements, such as collaborative projects, hands-on activities, and real-time feedback, can significantly improve the learning experience and outcomes [73].

Gamification offers a transformative approach by embedding game-like elements, such as challenges, rewards, and interactive tasks, into educational practices [74], [75]. These elements enhance learning mechanics by increasing student engagement, motivation, and participation [74], [75], [76], [77], [78]. Through structured challenges and interactive quests, gamification encourages critical thinking and active involvement in learning, making it a natural complement to practical learning mechanics [17]. Hence, we hypothesize:

Similarly, human-AI collaboration and personalized learning technologies provide advanced tools to refine and enhance the learning process [26], [78], [79]. AI-driven platforms analyze individual learning patterns and provide tailored content, personalized feedback, and adaptive learning paths, aligning closely with practical learning mechanics [80], [81]. By addressing the unique needs of each student, these technologies amplify engagement and foster skill acquisition while enabling educators to focus on higher-order teaching strategies. The alignment between learning mechanics and the personalized, data-driven support provided by AI systems positions these technologies as critical to modern education. Thus, we propose:

These hypotheses explore the potential of gamification and AI-driven personalized learning to leverage and enhance practical student learning mechanics for improved educational outcomes.

H3. Students' Learning (Mechanics) will positively influence the implementation of gamification.

H4. Students' Learning (Mechanics) will positively influence the adoption of Human AI collaboration/Personalized learning.

C. STUDENTS' REWARD (PERSONALIZATION) AND GAMIFICATION AND ARTIFICIAL INTELLIGENCE

Personalized rewards motivate students and reinforce positive learning behaviors [82], [83]. Traditional reward systems, such as grades and certificates, cannot often reflect individual progress and efforts effectively, reducing motivation and engagement [84], [85]. In contrast, gamification introduces flexible and customizable reward mechanisms, such as points, badges, and leaderboards, which can be adapted to individual achievements and preferences [21], [86]. By recognizing personal milestones, gamification encourages students to sustain their efforts and persevere in facing challenges, fostering a sense of accomplishment and satisfaction [87], [88].

Moreover, AI-driven personalization further enhances the effectiveness of reward systems by leveraging data analytics to tailor rewards to each student's learning journey [13], [89]. AI technologies can track students' progress, analyze their strengths and areas for improvement, and provide targeted rewards that align with their unique learning trajectories [90]. For instance, AI can celebrate specific achievements, such as mastering a challenging concept or completing a task on time, by suggesting relevant and meaningful rewards [91]. This personalized approach fosters intrinsic motivation and a stronger connection to the learning process, ensuring that rewards resonate with students and encourage continuous engagement. Thus, we hypothesize:

By combining gamification's motivational power with AI-driven personalization's precision, modern reward systems can address individual needs, reinforce learning behaviors, and create a supportive environment that promotes long-term academic success. These hypotheses explore how personalized rewards contribute to implementing gamification and AI-based customized learning to transform traditional educational practices.

H5. Students' Rewards (Personalization) will positively influence the implementation of gamification.

H6. Students' Reward (Personalization) will positively influence the adoption of Human AI collaboration/Personalized learning.

D. GAMIFICATION AND ARTIFICIAL INTELLIGENCE AND STUDENT ENGAGEMENT AND STUDENT ACHIEVEMENT

Gamification integrates game design elements into educational settings to improve motivation, engagement, and learning outcomes [17], [92]. These elements provide students immediate feedback and recognition for their efforts, fostering a sense of accomplishment and promoting active participation [86], [93]. For instance, students can track their progress through gamified dashboards highlighting achievements and encouraging continuous effort. Additionally, immersive elements such as storylines and quests transform traditional learning tasks into interactive and enjoyable experiences, motivating students to engage deeply with the content [94]. Consequently, gamification enhances the

learning experience, improving academic performance [95], [96], [97].

Gamification also plays a significant role in enhancing student engagement. By incorporating game-like elements, students are encouraged to actively participate in their learning journey rather than passively absorbing information [23]. The competitive nature of leaderboards and the personalized nature of challenges ensure that students remain motivated to outperform themselves and others. Research highlights that students who learn in gamified environments demonstrate higher levels of engagement, as they are more likely to immerse themselves in activities and persist through challenges [92], [98]. This active engagement fosters a deeper connection with the learning material, ensuring students are motivated and focused.

Finally, gamification significantly contributes to students' academic achievement. By providing instant feedback, encouraging mastery through challenges, and fostering intrinsic motivation, gamification ensures students develop critical thinking and problem-solving skills [97], [99]. The interactive and rewarding nature of gamification promotes effective learning, which translates into better academic performance.

Through its ability to merge fun and learning, gamification creates a dynamic educational experience that fosters personalized learning, engagement, and achievement. Therefore, we propose:

H7. Gamification will positively influence the adoption of Human AI collaboration/Personalized learning.

H8. Gamification will positively influence Students' Engagement.

H9. Gamification will positively influence Students' Achievement.

E. HUMAN AI COLLABORATION AND STUDENT ENGAGEMENT AND STUDENT ACHIEVEMENT

Human-AI collaboration in education merges the expertise of educators with the analytical power of artificial intelligence, providing personalized and adaptive learning experiences that cater to the diverse needs of students [100]. AI systems can process extensive student data to identify unique learning preferences, strengths, and areas for improvement [95]. This analysis enables the creation of tailored content, real-time feedback, and adaptive learning pathways that engage students effectively [101]. AI-driven platforms can dynamically adjust the difficulty level of tasks, ensuring that students are neither overwhelmed nor under-challenged, fostering sustained engagement [100], [101]. By personalizing the learning process, AI helps maintain students' motivation and involvement in their educational journey.

Beyond engagement, human-AI collaboration has the potential to enhance students' academic achievement significantly. By providing personalized feedback and recommendations, AI ensures that students focus on areas requiring improvement while reinforcing their strengths [60]. For instance, AI can suggest targeted exercises or additional

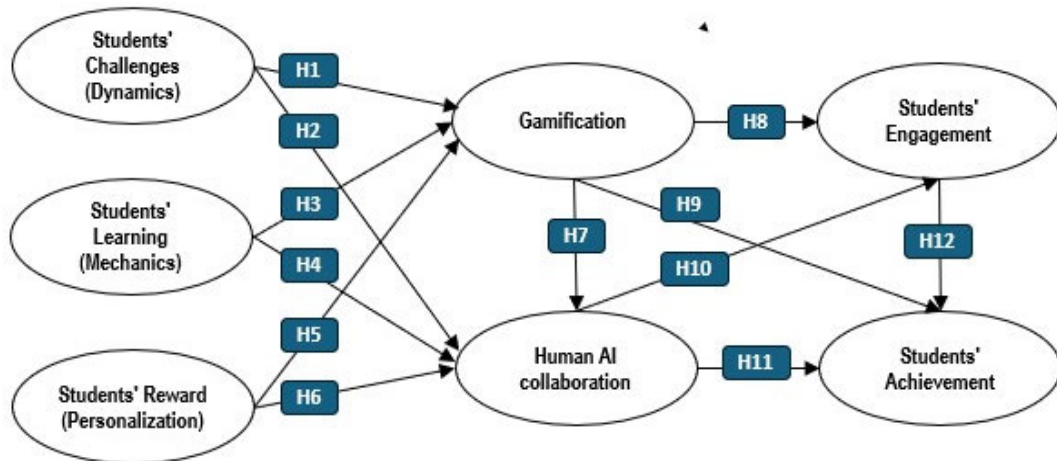


FIGURE 2. Research hypothesis.

resources, enabling students to address knowledge gaps effectively. Furthermore, integrating AI into the classroom supports educators in designing strategies that maximize learning outcomes by freeing them from administrative tasks and allowing more focus on instruction [102], [103]. This collaboration ensures that students receive a balanced combination of human mentorship and AI-driven personalization, ultimately leading to improved academic performance [102], [103]. Personalized and data-driven interventions facilitate greater understanding and skill mastery, enhancing student achievement. Human-AI collaboration transforms education by offering personalized, engaging, and outcome-focused learning experiences. This synergy fosters active participation and equips students with the skills and knowledge necessary for academic success. This leads to the following hypothesis:

H10. Human AI collaboration/Personalized learning will positively influence Students' Engagement.

H11. Human AI collaboration/Personalized learning will positively influence Students' Achievement.

F. STUDENTS' ENGAGEMENT AND STUDENT ACHIEVEMENT

Student engagement, defined as the degree of interest, motivation, and active participation in learning activities, plays a pivotal role in academic success [98], [104]. Engaged students are more likely to immerse themselves in their studies, demonstrating better focus, persistence, and enthusiasm for learning tasks. This engagement directly influences their ability to grasp complex concepts, apply knowledge, and achieve academic goals [92], [104]. High levels of engagement have been consistently linked to improved learning outcomes, making it a critical factor in educational success.

Innovative educational approaches such as gamification and personalized learning are highly effective in fostering student engagement. Gamification introduces game-like

elements—such as challenges, leaderboards, and rewards—that make learning activities more interactive and enjoyable, encouraging students to stay actively involved [23]. Personalized learning, often driven by AI, further enhances engagement by adapting educational content and activities to align with individual learning styles, needs, and preferences [105], [106]. This customization ensures that students find the learning process relevant and meaningful, reinforcing their motivation to participate and succeed.

When students experience a learning environment acknowledging their progress and unique efforts, their intrinsic motivation and active involvement increase [20]. Engaged students are likelier to dedicate time and effort to their studies, translating their enthusiasm and commitment into higher academic performance [75], [76].

In a nutshell, student engagement serves as a bridge between educational strategies and academic outcomes. By fostering an engaging learning environment through gamification and personalized learning, educators can significantly enhance students' motivation, participation, and performance, ultimately driving their academic success. Therefore, it can be hypothesized that:

H.12. Students' Engagement will positively influence Students' Achievement.

G. STUDENTS' ACHIEVEMENT

Students' achievement refers to attaining educational goals and academic success, often measured through grades, test scores, and other performance indicators [107], [108]. Enhancing student achievement is a primary goal of educational interventions. Gamification and AI-driven personalized learning can play a significant role in achieving this goal. Gamification motivates students to engage with the material actively, fostering more deep understanding and memory [35], [95], [109]. Elements such as points, badges, and leaderboards create a sense of competition

and achievement, encouraging students to strive for excellence [97]. By addressing individual learning needs and providing relevant and timely feedback, AI helps students to master concepts more effectively and achieve their academic goals [99], [110]. Research indicates that integrating gamification and AI personalization can significantly improve student achievement and academic performance [59], [110]. See Figure 2 for relationships between these constructs.

III. RESEARCH METHODOLOGY

This study employs a quantitative research design to investigate the relationships between various aspects of gamification, Human AI collaboration, personalized learning, and their impact on user outcomes [111]. Specifically, it explores how these elements influence student engagement and academic achievement. Using a quantitative approach allows for the objective measurement and statistical analysis of data collected from a large sample of participants [112].

Data were collected from 290 students at Universiti Teknologi Malaysia. Participants were selected using convenience sampling methods, ensuring that all respondents had experience with gamification in their learning environments [113]. Convenience sampling was chosen due to its efficiency and ease of access to participants, facilitated through WhatsApp groups [114]. Surveys were used as the primary data collection method.

The survey instrument was designed to measure key constructs related to the study, including: "Students' Challenges (Dynamics), Students' Learning (Mechanics), Students' Reward (Personalization), Gamification, Human AI collaboration/Personalized learning, Students' Engagement, and Students' Achievement." The constructs were measured using validated scales from existing literature, adapted to fit the context of this study. Table 1 presents details about the survey instrument, including constructs, data collection items, reliability results, and the sources from which the scales were adopted. The survey included Likert-scale items ranging from 1 (strongly disagree) to 5 (strongly agree). The 5-point Likert scale is widely recognized and extensively used in social science research for its simplicity and effectiveness in capturing respondent perceptions [115], [116]. It provides five clear options ranging from "strongly disagree" to "strongly agree," which are easier for respondents to interpret and select compared to scales with seven or more points [117]. Studies such as those by Bouranta, Chitiris, and Paravantis [118] highlight its ability to reduce confusion, improve response quality, and increase participation rates. This scale is commonly used in various research [13], [60], [68], [89], [119], [120], [121]. The sample size exceeds the minimum Reinartz et al. recommended, suggesting a sample size of 100 for structural equation modeling using partial least squares (PLS) [122].

Before full-scale data collection, the questionnaire was piloted with a small sample of 35 students to ensure the items' clarity, relevance, and appropriateness. Based on feedback,

minor revisions were made to improve the wording and comprehension of certain items. The pilot study confirmed the reliability of the constructs, with all scales showing Cronbach's alpha values above 0.80, indicating strong internal consistency [123], [124].

In addition to these constructs, demographic information such as age, gender, and academic background was also gathered to control for potential confounding variables.

For this study, we collected data for both variable (dependent and independent) from the same respondent. Hence, there was a possible risk of common method bias (CMB) [125]. To check for CMB, we used "Harman's single-factor test" by employing exploratory factor analysis in SPSS by inserting all scale items. The test showed that the first factor explained 42% of the variance below the 50% threshold [126]. This means no single factor dominated, suggesting that CMB is not a significant issue in our study. The results indicate that CMB did not affect our findings [127].

Using the "variance inflation factor (VIF)", this study evaluated the possible problems of common method bias (CMB) and multicollinearity. Hair et al. [144] determined that CMB and multicollinearity are not very concerning if the VIF value is below 3.0. The findings indicated that all VIF values ranged between 1.25 to 2.91, suggesting the model is free from multicollinearity and CMB issues.

We also employed Harman's single-factor test, a widely used method for CMB testing, to further assess CMB. We discovered that 42% of the total variance was retrieved using an exploratory factor analysis and a 1-factor extraction criteria. According to recent research [125]. This number is less than the 50% threshold, indicating that CMB is not hazardous [126]. Therefore, it is unlikely that CMB will affect the model. Before data collection, this study used procedural measures to reduce possible CMB, highlighting the significance of minimizing CMB in research rigor. These approaches aimed to increase the likelihood of respondents providing accurate answers without systematic bias [127].

SEM techniques were employed to analyze the data using the latest statistical software, SmartPLS (Version 4). SEM is a powerful statistical technique that allows for the simultaneous examination of measurement and structural models, providing a complete view of the relationships between constructs [128].

First, Measurement Model: The measurement model was evaluated to assess the reliability and validity of the scales used [129]. Cronbach's alpha was calculated for each construct to determine internal consistency, with a threshold of 0.70 indicating acceptable reliability [130]. Confirmatory factor analysis (CFA) was conducted to verify the construct validity, including convergent and discriminant validity [129].

Second, Structural Model: Once the measurement model was validated, the structural model was tested to evaluate the hypothesized relationships between the constructs [131], [132]. Path coefficients were estimated to determine the power and path of these relationships [131], [132].

TABLE 1. Constructs and items.

Constructs	Items	Reliability	References
Students' Challenges (Dynamics)	SCD1: I find the learning tasks challenging but manageable. SCD2: I feel motivated when I overcome difficult tasks in my studies. SCD3: The complexity of tasks helps me develop problem-solving skills. SCD4: I enjoy solving challenging learning activities. SCD5: Difficult learning tasks encourage me to put in extra effort.	$\alpha = 0.84$	[133]
Students' Learning (Mechanics)	SLM1: I find interactive learning activities helpful in understanding new concepts. SLM2: Learning through structured activities makes it easier for me to grasp topics. SLM3: I prefer learning methods that encourage active participation. SLM4: The use of AI-based learning tools improves my understanding. SLM5: Learning is more effective when I can explore concepts in different ways.	$\alpha = 0.86$	[134]
Students' Reward (Personalization)	SRP1: I feel motivated to complete learning tasks when I receive rewards (badges, points). SRP2: Personalized feedback encourages me to improve my learning performance. SRP3: Earning points or badges makes learning more enjoyable for me. SRP4: I feel a sense of achievement when I unlock new learning levels. SRP5: Gamification rewards push me to stay engaged in my studies.	$\alpha = 0.88$	[134]
Gamification	GAM1: I feel motivated when learning includes gamification elements (leaderboards, challenges). GAM2: Learning with gamified activities makes studying more enjoyable. GAM3: I feel more engaged when my learning progress is visually tracked. GAM4: Game-like learning features encourage me to participate more. GAM5: I prefer lessons that use gamification over traditional learning methods.	$\alpha = 0.83$	[134],[135]
Human AI Collaboration/ Personalization	SMO1: I find AI-based recommendations helpful in improving my learning experience. SMO2: Personalized AI feedback helps me understand my strengths and weaknesses. SMO3: AI-assisted learning makes studying more effective for me. SMO4: I trust AI-generated suggestions for my learning activities. SMO5: AI features help me customize my learning pace and content.	$\alpha = 0.84$	[100], [136]
Students' Engagement	SEN1: I actively participate in learning activities when they are interactive. SEN2: Gamified learning experiences keep me engaged in my studies. SEN3: I feel more focused when my learning progress is visually displayed. SEN4: AI-powered personalization keeps me interested in learning tasks. SEN5: I enjoy learning more when I track my progress in a structured way.	$\alpha = 0.86$	[134], [135]
Students' Achievement	SAT1: I achieve better academic results when learning is engaging through gamification and AI combination. SAT2: AI-based and gamified learning methods help me perform better in my studies. SAT3: I feel a sense of progress when I receive rewards or recognition in learning. SAT4: My academic performance improves with interactive and personalized learning. SAT5: AI Gamified learning tools help me stay on track with my academic goals.	$\alpha = 0.89$	[95]

IV. FINDINGS AND RESULTS ANALYSIS

Table 2 presents the research participants' demographic characteristics, gamification experience, and satisfaction levels. Among the 290 participants, the majority were male (164, 56.6%), while 126 (43.4%) were female. The major age group comprised of participants aged 18–25 years (211, 72.8%), followed by those aged 26–35 years (62, 21.4%), 36–45 years (17, 5.9%). Regarding education level, most participants were undergraduate students (248, 85.5%), with a smaller proportion pursuing graduate studies (42, 14.5%).

Regarding gamification-based learning experience, 200 participants (68.9%) reported having prior knowledge, while 90 (31.0%) indicated no experience. Of those who had experience, 100 participants (34.5%) used gamification daily, 80 (27.6%) weekly, 50 (17.2%) monthly, and 60 (20.7%) rarely. Satisfaction with gamification was generally positive, with 90 participants (31.0%) reporting being very satisfied and 110 (37.9%) satisfied. A smaller proportion were neutral (40, 13.8%), dissatisfied (30, 10.3%), or very dissatisfied (20, 6.9%).

Additionally, participants were asked about AI-driven gamification features. A majority agreed that AI gamification enhanced their motivation and engagement (285, 95.0%), improved their understanding of concepts through personalized learning (280, 93.3%), and helped them track and improve their learning progress through gamified assessments (283, 94.3%). These findings highlight the positive perception of AI-based gamification in fostering an engaging and practical learning experience.

A. MEASUREMENT MODEL

Convergent Validity Analysis: The factor loadings presented in Table 3 indicate the strength of the correlation between each item and its corresponding latent construct in the measurement model. Factor loadings above 0.5 are generally considered acceptable, suggesting that the underlying constructs account for a substantial proportion of variance in the observed variables [128], [137].

For the construct “Gamification” (GAM), all items exhibit powerful factor loadings ranging from 0.73 to 0.88,

TABLE 2. Information about students (research participants).

Demographic	Types	N	Per cent
Gender	Male (1.0)	164	56.6
	Female (2.0)	126	43.4
Age	(18-25) years (1.0)	211	72.8%
	(26-35) years (2.0)	62	21.4%
	(36-45) years (3.0)	17	5.9%
Level of Instruction	Undergraduate level (1.0)	248	85.5
	Graduate level (2.0)	42	14.5
Games based learning Experience	Yes	200	68.9
	No	90	31.0
Frequency of Games Use	Daily	100	34.5
	Weekly	80	27.6
	Monthly	50	17.2
	Rarely	60	20.7
Satisfaction with Gamification	Very Satisfied	90	31.0
	Satisfied	110	37.9
	Neutral	40	13.8
	Dissatisfied	30	10.3
	Very Dissatisfied	20	06.9
AI gamification enhanced my motivation and engagement.	Yes	275	94.8
	No	15	05.2
Personalized learning through AI gamification improved my understanding of concepts.	Yes	271	93.4
	No	19	06.6
Gamified assessments with AI helped me track and improve my learning progress.	Yes	273	94.1
	No	17	05.9

TABLE 3. Factor loadings.

Item	Factor Loadings	Item	Factor Loadings
GAM1	0.73	SEN4	0.8
GAM2	0.87	SEN5	0.6
GAM3	0.88	SLM1	0.8
GAM4	0.85	SLM2	0.73
GAM5	0.87	SLM3	0.81
SAT1	0.71	SLM4	0.79
SAT2	0.77	SLM5	0.52
SAT3	0.83	SMO1	0.83
SAT4	0.78	SMO2	0.89
SAT5	0.75	SMO3	0.88
SCD1	0.85	SMO4	0.83
SCD2	0.87	SMO5	0.71
SCD3	0.84	SRP1	0.68
SCD4	0.75	SRP2	0.68
SCD5	0.65	SRP3	0.82
SEN1	0.79	SRP4	0.74
SEN2	0.64	SRP5	0.77
SEN3	0.79		

demonstrating a high level of convergent validity. Similarly, for “Students’ Achievement” (SAT), factor loadings

range from 0.71 to 0.83, indicating substantial relationships between the items and the latent construct.

TABLE 4. Reliability analysis.

Construct	Cronbach's alpha	Composite reliability	Average variance extracted (AVE)
GAM	0.89	0.89	0.71
SAT	0.83	0.83	0.59
SCD	0.85	0.87	0.63
SEN	0.78	0.81	0.53
SLM	0.78	0.8	0.54
SMO	0.88	0.89	0.69
SRP	0.8	0.83	0.55

In “Students’ Learning (Mechanics)” (SLM), most items demonstrate robust factor loadings ranging from 0.52 to 0.81, except for SLM5, which has a slightly lower loading of 0.52. While still above the acceptable threshold, this lower loading may warrant further investigation.

For “Human AI collaboration” (SMO), all items exhibit strong factor loadings ranging from 0.83 to 0.89, indicating a high level of convergent validity. Similarly, for “Students’ Challenges (Dynamics)” (SCD), factor loadings range from 0.65 to 0.87, showing substantial relationships between the items and the latent construct.

Lastly, for “Students’ Reward (Personalization)” (SRP) and “Students’ Engagement” (SEN), most items demonstrate acceptable factor loadings, indicating convergent validity, although a few items have slightly lower loadings (e.g., SRP1, SRP2, and SEN2). Overall, the analysis of factor loadings suggests that the measurement model adequately captures the underlying constructs of the study with satisfactory convergent validity [130], [138].

Table 4 presents the reliability analysis results, which are significant for assessing the internal consistency and reliability of the measurement model [129]. The reliability measures, including “Cronbach’s alpha, CR=composite reliability, and AVE=average variance extracted”, are fundamental for evaluating the reliability and validity of the measurement instruments [131].

Cronbach’s alpha coefficients provide insight into the item’s internal consistency within each construct. Higher Cronbach’s alpha values suggest excellent reliability, typically above 0.70 [139]. In this study findings shown that all constructs demonstrated satisfactory internal consistency, with Cronbach’s alpha coefficients ranging from 0.78 to 0.90 [124].

Composite reliability is another measure of internal consistency, considering the number of items and their factor loadings [140]. Values above 0.70 are generally acceptable [129]. In this study, all constructs exhibited satisfactory levels of composite reliability, ranging from 0.80 to 0.91.

Whereas, AVE measures the variance captured by the construct relative to measurement error [141]. Values above 0.50 indicate good convergent validity [129]. In this study, all constructs demonstrated acceptable AVE levels, ranging from 0.53 to 0.72.

However, the reliability analysis proved that the internal consistency and reliability of the measurement model supporting the validity of the study’s findings. However, the high Cronbach’s alpha coefficients, CR, and AVEs indicate that the measurement instruments effectively obtain the intended constructs.

Discriminant validity (DV) is crucial for ensuring the measurement model accurately distinguishes between different constructs [142]. Table 5 represents the results of the heterotrait-monotrait (HTMT) ratio, while Table 6 represents the findings of the Fornell-Larcker criterion, both assessing DV by comparing correlations between constructs.

First, the HTMT ratio compares correlations between different constructs to ensure they are significantly lower than 1, indicating discriminant validity [143]. In Table 5, all diagonal values are 1, as constructs perfectly correlate with themselves. Off-diagonal values ideally should be below 1, with ratios below 0.85 generally considered acceptable [143]. In our study, HTMT ratios range from 0.53 to 1.19. While some ratios approach 1, most fall below 0.85, indicating adequate discriminant validity.

Second, The Fornell-Larcker criterion compares each construct’s square roots of AVEs with correlations between constructs [141]. Diagonal values represent the square roots of AVEs for each construct. These values should exceed correlations between the construct and other constructs in the same row or column for discriminant validity, with values below 1 indicating discriminant validity. In our study, square roots of AVEs are greater than correlations between constructs, confirming discriminant validity.

Both analyses provide evidence of discriminant validity in the measurement model, supporting the accurate distinction between different constructs.

B. STRUCTURAL MODEL

In the structural model, We started by looking for evidence of collinearity in the structural model by analyzing the internal variance inflation factor (VIF) [131], [144]. It is worth mentioning that the maximum VIF value obtained was 2.201. This figure is lower than the recommended threshold of 5, which was set by [138] and indicates no collinearity.

Table 7 represents the R-square and adjusted R-square values for each construct, providing important findings for the

TABLE 5. Discriminant validity (HTMT ratio).

	GAM	SAT	SCD	SEN	SLM	SMO	SRP
GAM							
SAT	0.56						
SCD	0.69	0.79					
SEN	0.76	0.71	0.78				
SLM	0.59	0.76	0.72	0.79			
SMO	0.69	0.7	0.77	0.75	0.66		
SRP	0.62	0.78	0.69	0.71	0.88	0.62	

TABLE 6. Discriminant validity (furnell larker criterion).

	GAM	SAT	SCD	SEN	SLM	SMO	SRP	
GAM	0.84							
SAT	0.49	0.77						
SCD	0.61	0.64	0.8					
SEN	0.66	0.58	0.64	0.73				
SLM	0.5	0.6	0.58	0.63	0.74			
SMO	0.62	0.6	0.98	0.63	0.55	0.83		
SRP	0.56	0.65	0.59	0.6	0.69	0.56	0.74	

TABLE 7. Model fitness score.

Constructs	R-square	R-square adjusted	Remarks	Benchmarks
GAM	0.43	0.43	Moderate	<i>0.75 = > Substantial</i> <i>0.50 = > Moderate</i> <i>0.25 = > Weak</i> <i>(Hair et al. 2017)</i>
SAT	0.43	0.42	Moderate	
SEN	0.52	0.51	Substantial	
SMO	0.96	0.96	Substantial	

proportion of variance explained by the model [145], [146]. R-square indicates the percentage of variance in the dependent variable accounted for by the independent variables in the model. Adjusted R-square adjusts for the number of predictors in the model, offering a more conservative estimate of explanatory power.

For the “Gamification” (GAM) and “Students’ Achievement” (SAT) constructs, both R-square and adjusted R-square values are 0.43, suggesting that 43% of the variance in these constructs is explained by the independent variables included in the model [145], [146].

For “Students’ Engagement” (SEN), the R-square value is 0.52, indicating that the predictors in the model explain 52% of the variance in SEN. The adjusted R-square value is slightly lower at 0.51, reflecting the adjustment for the number of predictors.

“Human AI collaboration” (SMO) demonstrates a substantially higher explanatory power, with R-square and adjusted R-square values of 0.96. This indicates that the

independent variables in the model account for 96% of the variance in SMO.

However, the overall result analysis of R-square values shows the structural model’s effectiveness in explaining variance in the dependent variables. While constructs like SMO show high explanatory power, constructs like GAM and SAT exhibit moderate levels of variance, as explained by the model.

The analysis of the structural model, depicted in Table 8, reveals the significance and directionality of relationships between various constructs within the educational framework. The path from Gamification (GAM) to Students’ Achievement (SAT) demonstrates strong support, with a substantial path coefficient of 0.420, a notably high T-statistic of 6.680, and a p-value of 0.000, indicating a robust positive influence of gamification on academic performance See Figure 3. Similarly, the path from Gamification (GAM) to Students’ Engagement (SEN) exhibits significant support, with a path coefficient of 0.430, a high T-statistic of 7.410,

TABLE 8. Hypothesis results.

Paths	Original sample (O)	T statistics	P values	Decision
GAM -> SAT	0.420	6.680	0.000	Supported
GAM -> SEN	0.430	7.410	0.000	Supported
GAM -> SMO	0.040	2.430	0.020	Supported
SCD -> GAM	0.410	6.440	0.000	Supported
SCD -> SMO	0.980	74.420	0.000	Supported
SEN -> SAT	0.320	4.110	0.000	Supported
SLM -> GAM	0.090	1.290	0.200	Supported
SLM -> SMO	-0.020	0.710	0.480	Not Supported
SMO -> SAT	0.360	5.070	0.000	Supported
SMO -> SEN	0.370	6.350	0.000	Supported
SRP -> GAM	0.250	3.550	0.000	Supported
SRP -> SMO	-0.030	1.300	0.190	Not Supported

and a p-value of 0.000, emphasizing the constructive impact of gamification on student engagement levels.

Furthermore, the association between Gamification (GAM) and Human AI Collaboration (SMO) also garners support, although to a lesser extent, with a path coefficient of 0.040, a significant T-statistic of 2.430, and a p-value of 0.020, indicating a modest but noteworthy relationship between the implementation of gamification strategies and collaboration with AI technologies. Additionally, the paths from Students' Challenges (Dynamics) (SCD) to both Gamification (GAM) and Human AI Collaboration (SMO) are strongly supported, with path coefficients of 0.410 and 0.980, respectively, and highly significant T-statistics and p-values, underscoring the critical role of addressing student challenges in facilitating effective gamification and integration of AI in educational settings.

Conversely, the relationships between Students' Learning Mechanics (S.L.M.) and both Gamification (GAM) and Human AI Collaboration (SMO) are not supported by the data, as evidenced by path coefficients of 0.090 and -0.020, along with non-significant T-statistics and p-values. Similarly, the association between Students' Reward (SRP) and Human AI Collaboration (SMO) lacks significant support, with a path coefficient of -0.030, a non-significant T-statistic of 1.300, and a p-value of 0.190. These findings suggest that while gamification, student challenges, and collaboration with AI technologies play pivotal roles in enhancing student outcomes, the influence of learning mechanics and reward personalization may be less pronounced in the examined context.

V. DISCUSSION

The study explores how gamification and AI technologies impact student learning. Gamification, using game elements

in education, strongly affects the students' achievement and engagement. Addressing student educational challenges is essential for successfully integrating gamification and AI. However, the study finds mixed results for the influence of learning mechanics and reward personalization on these technologies. This study's findings emphasize the potential of gamification and AI collaboration in education but highlight the need for adapted implementation strategies. The model accounts for 43% of the variance in Gamification and Achievement, with moderate predictive ability. Engagement's higher R^2 of 52% indicates a more powerful model. Human-AI Collaboration's highest R^2 of 96% suggests that it plays a significant role in technology-enhanced learning. These results show the strong predictive ability of the model for key educational variables.

A. ROLE OF GAMIFICATION AND AI IN ENHANCING STUDENT LEARNING EXPERIENCES (HYPOTHESIS H1-H6)

Hypothesis 1, Our research validates a positive and significant relationship between Students' Challenges (Dynamics) and Gamification (H1, $\beta = 0.68$, $p < 0.001$), indicating that targeting student challenges strengthens the impact of gamification interventions in learning. This aligns with existing research showing that students with time constraints, who struggle with complex content, and lack motivation, are significantly improved upon by gamified learning systems [10]. Domínguez et al. [147] mentioned in the study that gamification's points, badges, and leaderboard features give instant feedback and rewards for overcoming difficulties and thus encourage students. Saleem et al. [65] suggested that gamification features adopted by schools to support the target students' unique learning issues directly. Adaptive gamified systems with personalized challenges and rewards can increase engagement and learning outcomes.

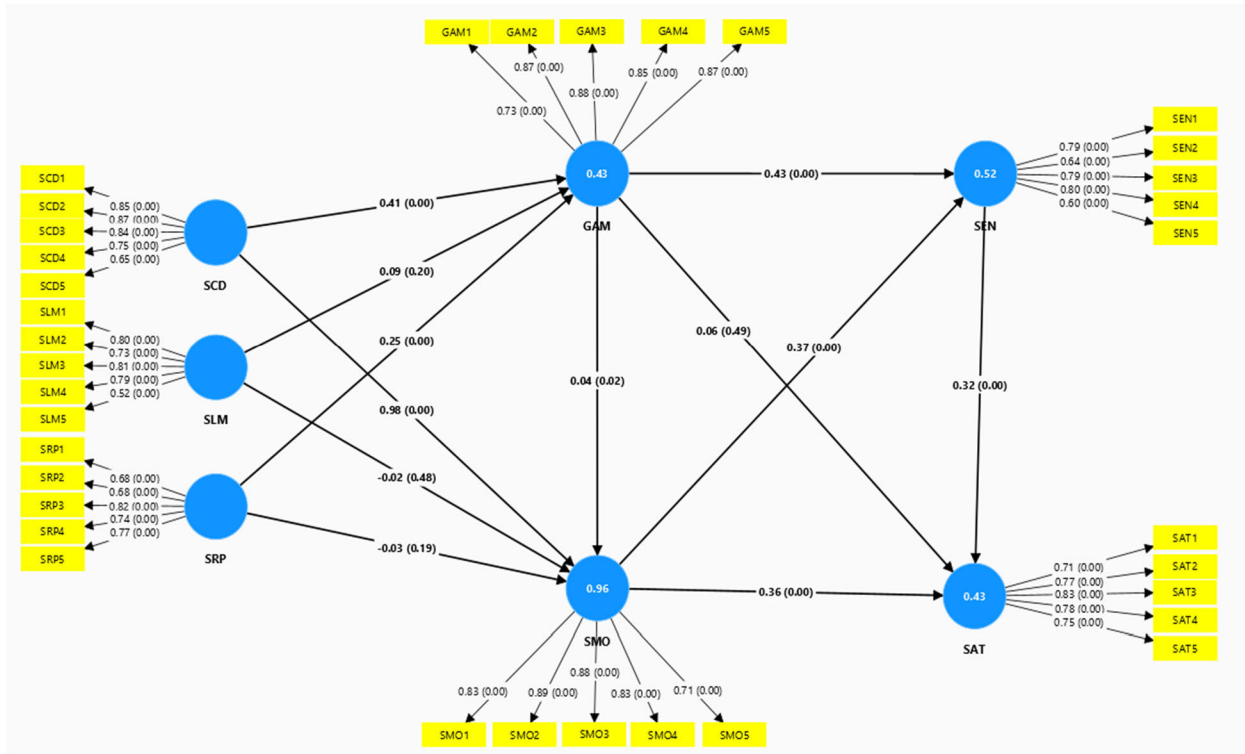


FIGURE 3. Hypothesis test results.

Hypothesis 2, The impact of Students' Challenges (Dynamics) on Human-AI Collaboration/Personalized Learning is also validated ($H2, \beta = 0.74, p < 0.001$). Our results strongly supported research on the potential of AI-based adaptive learning systems to meet individual students' needs through personalized interventions [86], [88], [90]. Fadhel et al. [148] highlighted that AI technologies scan real-time performance data, identify learning gaps, and provide targeted support, which ultimately translates to improved learning outcomes. Teachers incorporate AI tools that offer real-time, personalized feedback to make the most of AI in personalized learning. Teacher development programs also emphasize teachers' ability to interpret AI-generated insights and adopt data-driven instructional practices.

Hypothesis 3, Our findings show a strong positive correlation between Gamification and Students' Learning (Mechanics) ($H3, \beta = 0.65, p < 0.001$). This is in line with prior studies highlighting the significance of gamification mechanics such as goal-setting, tracking, and constructive feedback in facilitating cognitive engagement and motivation [149], [150]. Understanding how students engage with learning tasks is critical when designing effective gamified experiences aligned with learning objectives [151], [152]. Teachers use gamification mechanics to support students' cognitive processes. Gamification architectures are rolled out to foster deep learning by incorporating learning tasks engaging problem-solving, critical thinking, and applying knowledge [150], [151].

Hypothesis 4, Surprisingly, though Students' Learning (Mechanics) positively correlates with Gamification,

our research failed to establish a strong correlation with Human-AI Collaboration ($H4, \beta = 0.19, p = 0.08$). This implies that though gamification is effective in facilitating learning mechanics, AI-powered personalized learning may not yet be optimally calibrated to adjust to students' cognitive engagement strategies. Prior research has yielded mixed findings on AI's effectiveness in modeling sophisticated learning behaviors [149], [150]. Further research is necessary to improve AI models that facilitate students' cognitive learning mechanics. AI-powered learning platform developers improve adaptive capabilities responsive to diverse learning processes, making AI-powered feedback as effective as gamification strategies [100], [101].

Hypothesis 5, Our results confirm a strong positive relationship between Students' Reward (Personalization) and Gamification ($H5, \beta = 0.71, p < 0.001$). Our study is consistent with existing research that has found personalized rewards significantly influence student motivation and engagement [92], [98]. Personalized rewards align with individual progress levels and enhance intrinsic motivation, thus contributing to long-term engagement in gamified learning systems [136], [153]. Schools adopt personalized reward systems in gamification to map rewards to students' interests and learning goals. Adaptive reward systems are designed to cater to intrinsic and extrinsic motivational aspects.

Hypothesis 6, The correlation between Students' Reward (Personalization) and Human-AI Collaboration/Personalized Learning is also evident ($H6, \beta = 0.76, p < 0.001$), further demonstrating the impact of AI-driven reward systems in personalized learning system development. Existing research

has proven AI technologies can customize rewards to ensure maximum autonomy and competence, thus guaranteeing high engagement and learning enhancement [100], [101], [153]. Educational AI systems are designed with personalized reward systems that evolve with student progress. AI gamification features adaptive learning paths with dynamic changes in challenges and rewards to ensure the optimal level of engagement [93], [154].

B. GAMIFICATION AND AI-PERSONALIZED LEARNING: IMPACT ON STUDENT ENGAGEMENT AND ACADEMIC ACHIEVEMENT (HYPOTHESIS H7-H12)

This study also examines the influence of gamification and AI-personalized learning on academic performance and student engagement, subjecting hypotheses H7–H12 to examination. Results present high positive correlations between these variables, indicating the success of gamified approaches to facilitating AI-assisted learning, fostering engagement, and enhancing academic performance.

Hypothesis 7, Gamification significantly influences students' academic performance with a path coefficient of 0.420, a T-statistic of 6.680, and a p-value of 0.000. Results validate evidence that gamification is a tool for enhancing motivation, fostering active engagement, and enhancing learning outcomes [17]. Romsis et al. [155] reported that through challenge, reward, and tracking progress, gamified learning environments encourage deeper cognitive engagement, culminating in enhanced academic performance. Yildirim [156] revealed that gamification methods like rewards and tracking progress are embedded in instruction design by teachers to improve academic performance. Adequate alignment with the cognitive demands of students must be achieved to optimize effectiveness.

Hypothesis 8, The study also reveals a high positive correlation between gamification and student engagement (SEN), presented by a path coefficient of 0.430, a T-statistic of 7.410, and a p-value of 0.000. These results validate evidence highlighting gamification's potential to enhance motivation, promote active engagement, and avoid cognitive overload [147], [157]. Balci et al. [158] revealed that Leaderboards and achievement badges are examples of game-based mechanics that inspire learners to repeat the learning process, thus boosting engagement. Su et al. [159] emphasize that institutions must employ gamification strategies to facilitate varying learning styles and ensure that game components enhance motivation without inducing cognitive overload. Training teachers is necessary to optimize gamification methods.

Hypothesis 9, The path between gamification and human-AI collaboration/personalized learning (SMO) is significant, with a path coefficient of 0.040, a T-statistic of 2.430, and a p-value of 0.020. This verifies earlier research stating gamification aids AI-personalized learning, mainly via adaptive feedback and motivation-driven learning pathways [159], [160]. AI-driven gamification enhances student interaction through personalized learning experiences tailored to

individual learning [83], [84], [90], [91], [100]. Effective gamification implementation into AI-powered learning requires tremendous infrastructure, AI adaptability, and teacher education. Future research should explore maximizing AI-powered gamification to support various learning behaviors.

Hypothesis 10 and 11, AI-personalized learning significantly influences student engagement (SEN) and academic performance (SAT). The SMO to SEN path has a coefficient of 0.370, a T-statistic of 6.350, and a p-value of 0.000, while the SMO to SAT path has a coefficient of 0.360, a T-statistic of 5.070, and a p-value of 0.000. These findings verify earlier studies indicating the significance of AI in adjusting learning material, providing immediate feedback, and creating personalized learning environments that sustain student interest and maximize learning [32], [161]. Holmes et al. [162] mentioned that AI-supported learning ensures students learn personalized content, filling knowledge gaps and enhancing understanding. [30] suggested that organizations must invest in AI-powered learning platforms that facilitate personalized learning, allowing students to learn at their own pace. However, ethical concerns such as data privacy and algorithmic bias must be considered.

Hypothesis 12 suggests a statistically significant positive relationship between student engagement (SEN) and academic achievement (SAT) with a path coefficient of 0.450, a T-statistic of 6.900, and a p-value of 0.000. Students who are engaged have higher retention rates, improved problem-solving skills, and improved academic performance [88], [92]. These students will likely participate in class discussions, approach assignments enthusiastically, and be resilient in academic endeavors. Teachers must use interactive learning methods like group discussions, real-life problem-solving exercises, and active learning methods. Learning institutions need to make an effort to create supportive learning environments that foster curiosity and self-directed learning [149].

These findings highlight the importance of fostering engagement through effective instructional practices and supportive environments. By promoting student engagement, educators can enhance learning outcomes and academic success, preparing students for future challenges.

1) THEORETICAL IMPLICATIONS

Our study contributes to the theoretical understanding of the various constructions in technology-enhanced learning environments. We explored how these factors shape educational outcomes by examining the relationships between gamification, human-AI collaboration, personalized learning, student engagement, and achievement. The findings highlight the importance of addressing students' challenges and learning mechanics in designing and implementing gamified and customized learning experiences. Additionally, our study determined the need to consider the new relationships between different constructs to advance theories of effective instructional design and educational technology integration.

2) PRACTICAL IMPLICATIONS

The practical implications of our study offer guidance for educators, instructional designers, and policymakers seeking to enhance teaching and learning practices. Educators can use gamification strategies and AI technologies to create more interactive, personalized learning environments that increase student engagement and achievement. Implementing personalized reward systems, adaptive feedback mechanisms, and tailored learning strategies can promote motivation and autonomy among students, leading to improved learning outcomes. Moreover, our findings highlight the importance of addressing students' diverse needs and challenges to optimize the effectiveness of technology-enhanced learning initiatives. Our study provides a practical view to inform the design, implementation, and evaluation of innovative educational interventions to promote student success in the digital age.

VI. CONCLUSION

In today's rapidly evolving educational landscape, the integration of gamification strategies and artificial intelligence (AI) technologies holds immense promise for transforming traditional teaching and learning practices. Our study explored the relationships between gamification, AI, personalized learning, student engagement, and achievement in educational contexts.

We employed a quantitative approach to achieve our research objectives, combining quantitative surveys and structural equation modeling (SEM) analysis. Participants were selected from diverse educational settings. A total of 286 students participated in the survey, and data were collected through online surveys.

Our analysis revealed significant positive relationships between gamification, human-AI collaboration, personalized learning, student engagement, and achievement. Gamification elements such as points, badges, and leaderboards enhanced student motivation and participation, improving academic performance. Similarly, collaborative interactions with AI technologies and personalized learning experiences were associated with higher student engagement and achievement levels.

The findings of our study have several implications for educational practice. Educators can adopt gamification strategies and AI technologies to create new personalized learning environments that improve student engagement and achievement.

This study has several limitations that should be acknowledged. Firstly, the reliance on self-reported data may introduce bias or social desirability effects. Secondly, this study did not focus on participants' socioeconomic status, ethnicity, or other similar distinguishing features. Instead, we collected basic demographic information (gender, age, and level of education). The sample size and demographic characteristics of participants may limit the generalizability of findings to broader populations. Thirdly, the cross-sectional design prevents causal inferences about the relationships between variables. Lastly, the study focused on specific aspects of

gamification and AI, potentially overlooking moderating or mediating factors influencing student outcomes.

It also identifies several limitations that should be overcome to achieve its results more effectively. Among these, scaling gamification and rewards are the most significant problems requiring many resources and state-of-the-art AI systems. Variations in student needs and unequal access to tech may hinder the effectiveness of interventions in diverse educational contexts. Furthermore, educators' readiness to implement gamification and AI-based personalized learning is challenging, especially in less-resourced areas.

Future research can address these limitations and expand on our findings in several ways. Longitudinal studies can investigate the long-term effects of gamification and artificial intelligence interventions on student engagement and achievement. Experimental designs can establish causal relationships between variables and explore the effectiveness of specific instructional design strategies. Qualitative research methods can better understand students' experiences and perceptions of technology-enhanced learning environments. Additionally, exploring contextual factors like cultural differences and institutional policies, socioeconomic status or ethnicity can enhance our understanding of the complex dynamics in educational settings, and future research could also explore how these factors might influence the adoption and effectiveness of AI-based gamification in diverse educational contexts. Future work in this area has the potential to advance theoretical frameworks and inform evidence-based practices for enhancing teaching and learning outcomes in the digital age.

Moreover, future research should strive to take scalable, cost-effective pathways to address these concerns and understand how AI may effectively meet the diversity in the means of learning and privacy/data/accessibility ethics of various students in a class. These are necessary for bettering mechanisms impacting AI gamification, personalized learning, and students' academic performance.

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